

EE 172 – UNIT 3

Polyphase Induction Machines

Short Detailed Notes

1.1 INTRODUCTION TO POLYPHASE INDUCTION MOTORS

What is an Induction Motor?

- Most widely used motor in industrial and commercial applications
- Converts **electrical energy** to **mechanical energy**
- Operates on the principle of **electromagnetic induction**
- Reliable, robust, and relatively inexpensive

Two Main Types

Squirrel-Cage Motor	Wound-Rotor (Slip-Ring) Motor
Rotor has short-circuited bars	Rotor has 3-phase windings
More common	Less common
Low cost	Higher cost
High reliability	More complex
High efficiency	Lower efficiency
No slip rings/brushes	Has slip rings and brushes

1.1.2 Construction

Main Parts:

- 1 **Stator** – Stationary part
- 2 **Rotor** – Rotating part
- 3 **Air Gap** – 0.4 mm to 4 mm (depends on power rating)

1.1.2.1 Stator Construction

Components:

- **Steel frame** – encloses the core
- **Hollow cylindrical core** – made of stacked laminations
- **Slots** – punched uniformly around inner surface
- **3-phase winding** – placed in the slots

Core Laminations:

- Material: **Silicon steel**
- Thickness: 0.3 to 0.5 mm
- **Insulated** from each other
- Purpose: **Reduce eddy current loss**

1.1.2.2 Rotor Construction

General Features:

- Composed of punched laminations
- Slots for rotor winding
- Two types based on winding:

(a) Squirrel-Cage Rotor

- **Bare copper bars** pushed into slots
- Bars slightly longer than rotor
- Ends **welded** to copper end rings
- All bars **short-circuited**

For Small/Medium Motors ($\leq 1000\text{kW}$):

- Bars and end rings made of **die-cast aluminum**
- Form an **integral block**

Advantages:

- Simple and rugged
- No slip rings or brushes
- Explosion-proof (no sparking)

(b) Wound/Slip-Ring Rotor

- **3-phase windings** similar to stator
- Usually **star-connected**
- Terminals connected to **3 slip rings**
- Slip rings and brushes allow **external resistors** in rotor circuit

Advantage: Can add external resistance for:

- High starting torque
- Speed control

1.1.3 Principle of Operation

Step-by-Step Process:

1. Rotating Magnetic Field

- 3-phase voltage applied to stator
- Produces a **rotating magnetic field**

2. Voltage Induced in Rotor

- Rotating field **cuts across** rotor conductors
- Induces voltage in all rotor conductors

3. Current Flow

- Induced voltage causes **large currents** in rotor
- Rotor is short-circuited (or has external resistance)

4. Torque Production

- Current interacts with magnetic field
- Produces **mechanical torque**

5. Direction of Rotation

- According to **Lenz's Law**
- Rotor drags in the **direction** of revolving field
- Induced current opposes the change causing it

1.1.4 Slip

What is Slip?

- The rotor **never catches up** with synchronous speed
- If rotor reached synchronous speed: relative velocity = zero, no induced voltage, no current, no torque, rotor would slow down

Definition:

$$s = (N_s - N_r) / N_s$$

Where: N_s = Synchronous speed (rpm), N_r = Rotor speed (rpm)

Typical Slip Values:

Condition	Slip (s)
Standstill/Locked rotor	1.0
No load	< 0.001 (0.1%)
Full load (large motors >1000kW)	< 0.005
Full load (small motors ≤10kW)	< 0.03

Why Induction Motors are “Constant Speed”

- Slip is very small (1–5%)
- Speed changes very little with load

1.1.5 Speed of MMF Wave

Synchronous Speed Formulas:

$$\omega_s = \omega / p \text{ (rad/s)}$$

$$N_s = 60f / p \text{ (rpm)}$$

Where: ω = angular frequency of source (rad/s), f = frequency of source (Hz), p = number of pole-pairs

Note: One pole-pair = 2 poles

1.1.6 Rotor Frequency

Formula:

$$f_r = s \times f$$

Where: f_r = frequency of voltage induced in rotor, f = supply frequency, s = slip

Key Points:

- At standstill ($s=1$): $f_r = f$
- At full load ($s \approx 0.03$): $f_r \approx 1.5$ Hz (for 50 Hz supply)
- Rotor iron loss is **negligible** because f_r is very small

Example 1 – Calculating Slip

Problem: 3-phase, 50Hz motor, full-load speed = 1440 rpm

Synchronous speed near 1440 rpm = 1500 rpm (4-pole); $p = 2$ pole-pairs (4 poles)

$$N_s = 60 \times 50 / 2 = 1500 \text{ rpm}$$

$$s = (1500 - 1440) / 1500 = 0.04 \text{ (4\%)}$$

1.1.7 Power Flow in Induction Motor

Power Flow Diagram:

```
Input Power (P_e)
↓
- Stator Copper Loss (P_js)
- Stator Iron Loss (P_fs)
↓
Power transferred to Rotor (P_r)
↓
- Rotor Copper Loss (P_jr)
- Rotor Iron Loss (negligible)
↓
Gross Mechanical Power (P_m)
↓
- Friction & Windage Loss (P_v)
↓
Shaft Output Power (P_L)
```

Key Power Relations:

```
P_js = 3 × I12 × R1 (Stator copper loss)
P_r = P_e - P_js - P_fs (Power to rotor)

P_jr = s × P_r (Rotor copper loss)
P_m = P_r - P_jr = (1-s) × P_r (Gross mechanical power)
P_L = P_m - P_v (Shaft output power)
```

Torque Relations:

```
Gross Torque = P_m / ω_r
Shaft Torque = P_L / ω_r
Loss Torque = P_v / ω_r
```

Efficiency:

```
η = P_L / P_e
```

Example 2 – Power Calculations

Problem: $P_e = 40\text{kW}$, Stator losses = 1.5kW , $s = 0.04$, Friction loss = 0.8kW

(a) Total mechanical power developed:

```
P_r = 40 - 1.5 = 38.5 kW
P_jr = s × P_r = 0.04 × 38.5 = 1.54 kW
P_m = P_r - P_jr = 38.5 - 1.54 = 36.96 kW
```

(b) Output power:

```
P_L = P_m - P_v = 36.96 - 0.8 = 36.16 kW
```

(c) Efficiency:

```
η = 36.16 / 40 = 0.904 = 90.4%
```

Example 3 – Effect of Adding Rotor Resistance

Problem: Speed reduced to 40% of synchronous speed, torque unchanged

$$s = (100\% - 40\%) / 100\% = 0.6$$

$$P_r = 38.5 \text{ kW (torque unchanged)}$$

$$P_{jr} = s \times P_r = 0.6 \times 38.5 = 23.1 \text{ kW}$$

$$\text{Total rotor losses} = P_{jr} + P_v = 23.1 + 0.8 = 23.9 \text{ kW}$$

$$\text{Output power} = 38.5 - 23.9 = 14.6 \text{ kW}$$

$$\eta = 14.6 / (38.5 + 1.5) = 0.365 = 36.5\%$$

Key Insight: Adding rotor resistance increases slip and rotor losses, reducing efficiency!

1.1.8 Equivalent Circuit

Exact Equivalent Circuit (Per Phase)

Rotor Circuit Transformation:

- Actual rotating rotor → Fictitious stationary rotor
- Variable frequency → Fixed frequency (supply frequency)
- Variable reactance → Constant reactance
- Variable resistance → R_2'/s

Why This Works:

- Same current flows
- Same mmf wave produced
- Allows transformer-like analysis

Equivalent Circuit Components:

Component	Description
V_1	Stator phase voltage
R_1	Stator resistance
X_1	Stator leakage reactance
R_c	Core loss resistance
X_m	Magnetizing reactance
R_2'	Rotor resistance (referred to stator)
X_2'	Rotor leakage reactance (referred to stator)
R_2'/s	Effective rotor resistance (includes mechanical load)

Approximate Circuit (Neglect Magnetizing Branch)

- Valid for motors > 5 hp (3 kW)
- Error < 10%

- Simplifies calculations significantly

1.1.9 Torque Equations

General Torque Formula:

$$T_e = (3 \times I_2'^2 \times R_2'/s) / \omega_s$$

In Terms of Circuit Parameters:

$$T_e = (3 / \omega_s) \times [V_1^2 \times (R_2'/s)] / [(R_1 + R_2'/s)^2 + (X_1 + X_2')^2]$$

Torque-Slip Curve Characteristics

Key Features:

- 1 **Starting torque** (s=1)
- 2 **Maximum torque** (breakdown torque) at s = s_m
- 3 **Running torque** (normal operating region, s small)

1.1.9.1 Starting Torque

At s = 1:

$$T_e = (3 / \omega_s) \times V_1^2 \times R_2' / [(R_1 + R_2')^2 + (X_1 + X_2')^2]$$

Quadratic Nature:

- For a given starting torque, there are **two possible rotor resistances**
- One low, one high

1.1.9.2 Maximum (Breakdown) Torque

Slip at Maximum Torque:

$$s_m = R_2' / \sqrt{R_1^2 + X^2}$$

Where $X = X_1 + X_2'$

Maximum Torque:

$$T_{max} = (3 / 2\omega_s) \times V_1^2 / [R_1 + \sqrt{R_1^2 + X^2}]$$

Key Observation:

- $s_m \propto R_2'$ (increases with rotor resistance)
- T_{max} **independent of R_2'** (not affected by rotor resistance)

What This Means: Increasing rotor resistance:

- Shifts maximum torque to higher slip
- Doesn't change maximum torque value
- Useful for starting (higher starting torque)

1.1.9.3 Effect of Changing Rotor Resistance

Starting Torque vs Rotor Resistance

- As R_2' increases: starting torque **increases** initially

- Reaches maximum when $R_2' = \sqrt{R_1^2 + X^2}$
- Then **decreases**

Trade-off:

High Rotor Resistance	Low Rotor Resistance
High starting torque	Low starting torque
Speed drops with load	Good speed regulation
High rotor losses	Low rotor losses
Low efficiency	High efficiency

1.1.10 Comparison: Cage vs Slip-Ring Motors

Feature	Squirrel-Cage	Slip-Ring
Cost	Cheaper	More expensive
Robustness	More robust	Less robust
Efficiency	Higher	Lower
Power Factor	Better	Worse
Starting Torque	Lower	Much higher
Starting Current	Higher	Lower
Speed Control	Difficult	Easy (with external resistance)
Explosion-proof	Yes (no sparking)	No (brushes spark)

Example 4 – Complete Motor Analysis

Problem: 415V, 3-phase, 6-pole, 50Hz, star-connected slip-ring motor. $R_1 = 0.04\Omega$, $X_1 = 0.15\Omega$, $R_2' = 0.05\Omega$, $X_2' = 0.15\Omega$

(a) Stator current and gross torque at $s = 0.05$:

Synchronous speed:

$$\omega_s = 2\pi \times 50 / 3 = 104.7 \text{ rad/s}$$

Total impedance at $s=0.05$:

$$\begin{aligned} Z &= (R_1 + R_2'/s) + j(X_1 + X_2') \\ &= (0.04 + 0.05/0.05) + j(0.15 + 0.15) \\ &= 1.04 + j0.30 = 1.082\angle 16.09^\circ \Omega \end{aligned}$$

Phase voltage:

$$V_{ph} = 415/\sqrt{3} = 240V$$

Stator current:

$$I = 240\angle 0^\circ / 1.082\angle 16.09^\circ = 222\angle -16.09^\circ \text{ A}$$

Gross torque:

$$\begin{aligned} T_e &= 3 \times I^2 \times (R_2'/s) / \omega_s \\ &= 3 \times 222^2 \times (0.05/0.05) / 104.7 \\ &= 3 \times 49284 \times 1 / 104.7 \\ &= 1412 \text{ Nm} \end{aligned}$$

(b) Maximum torque and slip:

Slip at maximum:

$$\begin{aligned} s_m &= R_2' / \sqrt{(R_1^2 + X^2)} \\ &= 0.05 / \sqrt{(0.04^2 + 0.30^2)} \\ &= 0.05 / 0.3027 \\ &= 0.165 \text{ p.u.} \end{aligned}$$

Maximum torque:

$$\begin{aligned} T_{max} &= (3/2\omega_s) \times V_{ph}^2 / [R_1 + \sqrt{(R_1^2 + X^2)}] \\ &= (3/(2 \times 104.7)) \times 240^2 / [0.04 + 0.3027] \\ &= 0.01433 \times 57600 / 0.3427 \\ &= 2410 \text{ Nm} \end{aligned}$$

Motor speed at max torque:

$$\omega_r = \omega_s(1 - s_m) = 104.7(1 - 0.165) = 87.42 \text{ rad/s}$$

Output power at max torque:

$$P_o = T_{max} \times \omega_r = 2410 \times 87.42 = 210.7 \text{ kW}$$

(c) External resistance for maximum torque at standstill:

For maximum torque at $s=1$:

$$s_m = 1 = R_2' / \sqrt{R_1^2 + X^2}$$
$$R_2' = \sqrt{R_1^2 + X^2} = \sqrt{(0.04^2 + 0.30^2)} = 0.303\Omega$$

Additional resistance required:

$$R_{ext} = R_2' - 0.05 = 0.253\Omega \text{ (per phase, referred to stator)}$$

1.1.11 STARTING METHODS

1.1.11.1 Squirrel-Cage Motors

Two Methods:

- 1 Direct-on-line (DOL) starting
- 2 Reduced voltage starting

(a) Direct-on-Line (DOL) Starting

What It Is:

- Motor connected **directly** to full supply voltage
- Simplest and cheapest method

Characteristics:

- Starting current: **5–6×** full-load current
- Starting torque: **Full** torque
- Low power factor at start

Problems:

- Voltage dips in supply
- Affects sensitive equipment: incandescent lamps (dimming), television sets (flicker), precision machine tools

Supply Authority Limits:

- Depends on supply capacity
- May be 5kW or up to 75kW (on 415V)
- Large systems can start motors up to 7500kW DOL!

Protection:

- Motor protective circuit breaker (Q1)
- Overload protection
- Contactor with auxiliary contacts

(b) Reduced Voltage Starting

General Effect: For voltage reduced to fraction **x** of normal:

Starting Current: $I_s = x \times I_{sc}$

Starting Torque: $T_s = x^2 \times T_{sc}$

(i) Stator Resistance Starting

- Resistors in series with stator
- Voltage drop: high initially, decreases as motor speeds up
- Short-circuited at ~80% speed

Applications:

- Smooth starting of small machines

- Centrifugal oil purifiers

Advantages:

- Smooth acceleration
- Minimal shock when full voltage applied

Disadvantages:

- Resistors can burn out with frequent starting
- Energy loss in resistors

(ii) Autotransformer Starting

- Uses autotransformers to reduce voltage
- Common tap settings: 50%, 65%, 80%

Current Relations:

Motor current: $I_s = x \times I_{sc}$
 Line current: $I_L = x^2 \times I_{sc}$ (lower than resistance method!)
 Torque: $T_s = x^2 \times T_{sc}$

Comparison with Resistance Starting:

- **Lower line current** for same torque
- Autotransformers are more expensive

Disadvantages:

- Not smooth transition
- Transient current surge during changeover
- Motor momentarily disconnected

Construction:

- Usually two single-phase autotransformers (open delta)
- Three-phase transformer for maximum torque

(iii) Star-Delta Starting

- Motor designed for delta running
- Six leads brought out to terminal box
- Start in star, switch to delta

Voltage Reduction:

$x = 1/\sqrt{3} = 0.577$
 Starting Torque: $T_s = T_{sc}/3$
 Starting Current: $I_s = I_{sc}/3$

Advantages:

- **Cheapest** method
- Effective if starting torque adequate

Disadvantages:

- Not suitable >3kV
- Same transient issue as autotransformer
- Motor disconnected briefly during changeover

Timer Setting:

- Around 10 seconds
- Less for light loads
- More for heavy loads

Comparison of Starting Methods (Example 6):

Method	Starting Torque	Starting Current
DOL	1	1
Star-Delta	0.33	0.33
Autotransformer (40% tap)	0.16	0.16

1.1.11.2 Slip-Ring Motor Starting

Method:

- External resistors added to rotor circuit
- Resistance reduced in steps as motor accelerates

Advantages:

- **Reduced starting current**
- **Increased starting torque** (moves T_{max} to higher slip)
- Smooth acceleration
- Suitable for **heavy, frequent starting**

Why It Works:

- $s_m \propto R_2'$
- Adding resistance shifts maximum torque to starting point
- High starting torque with low starting current

KEY FORMULAS SUMMARY

Speed

$$\begin{aligned}N_s &= 60f/p \text{ (synchronous speed)} \\s &= (N_s - N_r)/N_s \text{ (slip)} \\f_r &= sf \text{ (rotor frequency)}\end{aligned}$$

Power

$$\begin{aligned}P_r &= P_e - P_{js} - P_{fs} \text{ (power to rotor)} \\P_{jr} &= sP_r \text{ (rotor copper loss)} \\P_m &= (1-s)P_r \text{ (mechanical power)} \\P_L &= P_m - P_v \text{ (output power)} \\\eta &= P_L/P_e \text{ (efficiency)}\end{aligned}$$

Torque

$$\begin{aligned}T_e &= (3I_2'^2 R_2'/s)/\omega_s \\T_e &= (3/\omega_s) \times V_1^2 (R_2'/s) / [(R_1 + R_2'/s)^2 + X^2] \\T_{\max} &= (3/2\omega_s) \times V_1^2 / [R_1 + \sqrt{(R_1^2 + X^2)}] \\s_m &= R_2' / \sqrt{(R_1^2 + X^2)}\end{aligned}$$

Starting (Reduced Voltage)

$$\begin{aligned}I_s &= x \times I_{sc} \\T_s &= x^2 \times T_{sc}\end{aligned}$$

IMPORTANT CONCEPTS TO REMEMBER

- 1 **Slip is essential** – without slip, no torque
- 2 **Rotor frequency** = $s \times$ supply frequency (small at normal operation)
- 3 **Rotor copper loss** = $s \times$ power transferred to rotor
- 4 **Maximum torque** independent of rotor resistance
- 5 **Slip at max torque** $\propto$ rotor resistance
- 6 **Cage motors** = simple, cheap, constant speed
- 7 **Slip-ring motors** = high starting torque, speed control
- 8 **Power factor correction** improves efficiency
- 9 **Induction motors are constant speed** machines (small slip)
- 10 **Starting current** 5–6 $\times$ full load current